

Le-Hoang Tran

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SUMMARY

Backend Software Engineer with nearly 5 years of experience building large-scale, high-performance systems in Java, Spring Boot, Apache Kafka, SQL/NoSQL and Kubernetes, processing 100M+ events daily for product serving 20M+ users.

TECHNICAL SKILLS

Languages: Java, Python, C, Javascript

Backend: Spring Framework, Spring Boot, Microservices, REST, gRPC, Camunda, JUnit, Mockito, Testcontainers

Infra & Data: MySQL, Apache Kafka, Redis, MongoDB, Elasticsearch, MinIO, ClickHouse

Platform & Ops: Kubernetes, Docker, Jenkins, Grafana, ELK, K6, Linux, Bash, Git

AI Tooling: Claude Code, OpenAI Codex, Agentic Workflows, Prompt Engineering

EXPERIENCE

LINE Technology Vietnam

Jun 2025 – Aug 2025

Backend Software Engineer

Hanoi, Vietnam

- Implemented an **online data migration solution** for **10TB+** of data from LINE and Yahoo systems to a unified MySQL database using dual-write and backfill strategy.
- Leveraged **AI-assisted development tools** and prompt engineering for source code onboarding, automated unit & integration tests, and early-stage code review, increasing development velocity by 30%.

Viettel Digital Finance

May 2021 – Jun 2025

Backend Software Engineer

Hanoi, Vietnam

Customer Engagement Platform

- Architected an **in-house marketing platform** to deliver personalized campaigns through push notifications, in-app messaging, and rewards distribution, processing **100M+** events daily.
- Increased user segmentation creation throughput from **500 to 35,000 QPS** while **reducing p95 latency by 80%** by introducing a **Roaring Bitmap**-based solution and multi-layer caching.
- Engineered a Camunda-based workflow orchestration module, enabling operations teams to configure and manage **200+** single- and multi-step marketing workflows per month.
- Led a team of 4 backend engineers** through Agile sprint planning, system design, and peer code reviews, guiding technical decisions to keep the system maintainable through rapid feature growth.
- Owned end-to-end system observability using Grafana for **20+** services, identifying performance bottlenecks and improving reliability through configuration tuning, caching strategies, and microservice patterns.

Data Tracking System

- Designed and delivered a high-throughput **in-house event tracking system** for mobile and web platforms, processing **150M+** events daily and replacing a third-party tracking provider to **reduce licensing costs**.
- Reduced event validation latency by **30%** by replacing linear rule processing with a decision-tree validation engine, supporting **3,000+** dynamic rules configurable online.
- Built non-blocking ingestion REST APIs using Spring WebFlux, sustaining **1,700+ RPS** on average while reducing request-per-thread overhead.

Smart Authentication System

- Collaborated on the research and delivery of a **Smart OTP** solution for **20M+** end users, eliminating reliance on SMS OTPs and saving over **\$200K/year**.
- Implemented secure key exchange, OTP generation, and digital signature mechanisms across critical APIs, meeting internal cryptographic and security-review requirements.
- Hardened Android client application using native **C/JNI**, applying obfuscation and root/jailbreak/tamper detection, successfully passing enterprise-level security audits.

EDUCATION

George Mason University

Aug 2025 – Dec 2026 (Expected)

M.S. in Computer Science, GPA: 3.86

Fairfax, VA, USA

Hanoi University of Science and Technology

Aug 2017 – May 2021

B.S. in Information Technology, GPA: 3.62 (top 5% of cohort)

Hanoi, Vietnam

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

Sep 2023

Oracle Certified Java SE Programmer

Apr 2023